

Design of a Two-Tier QoS-Aware Scheduler for Private 5G

A convex rate planner on a ~ 1 s horizon, over a per-slot drift-plus-penalty tracker.

Motivation, design, and simulation results against the scheduler OpenAirInterface ships — with what remains to land it on hardware.

The factory floor asks for three different things

One private 5G cell, one carrier, and traffic classes that differ in kind rather than in volume:

Class	Example	What it needs	How you know it failed
GBR	Robot camera and LIDAR uplink	A sustained rate floor	Dropped frames, blind robot
Delay	Motion control, teleoperation	Bytes before a deadline	A late command — a safety event
Best-effort	Firmware push, log upload	Whatever is left	Slower; nobody notices

OpenAirInterface's NR MAC schedules with a **proportional-fair metric over per-UE grants**. PF has no term for a rate contract, no term for a deadline, and no configured-grant path for periodic traffic. It is a good answer to a question the factory is not asking.

A robot whose video is switched off entirely is a failure in a way a uniformly degraded fleet is not — a distinction a throughput-fair metric cannot see, let alone act on.

What the prior art told us to do — and not to

Adopted

Two timescales. A slow declarative program over a fast per-slot controller — the pattern behind NVS (Kokku 2012) and the O-RAN RIC split. We instantiate it; we do not claim it.

Network utility maximisation (Kelly 1998) for the slow tier — an objective a plant engineer can read and audit.

Drift-plus-penalty (Neely 2010) for the fast tier — it solves the per-slot problem, and needs a target, which Tier-1 supplies.

Configured grants dominate latency for time-critical industrial traffic (Larrañaga 2023, ns-3 / 5G-LENA).

Ruled out, and why

Tuned PF variants — M-LWDF, Exp-rule. Excellent on delay; on rate they saturate, because a multiplicative deficit weight equalises a weighted $1/R_{avg}$ and cannot reach an arbitrary unequal target. Monghal (2008) measured exactly that.

Learned schedulers. Kela (2024), arguing from inside that literature, finds deployable deep schedulers do not yet exist — real-time cost, 3GPP compliance, and poor generalisation across bandwidth and traffic mix.

Two commitments follow: an **integrator, not a metric multiplier** in the fast tier, so an arbitrary target vector is reachable; and a **convex program with one auditable fairness dial** in the slow tier.

Tier-1: what rate should each flow target?

Decision variables are a target rate per flow and the shortfall slacks. All solves share one feasible set:

$$\sum_{i: d(i)=d} \frac{r_i}{SE_i} \leq C_d \quad \text{capacity in PRB-symbols, special slot included}$$

$$r_i + s_i \geq G_i \quad \text{GBR contract — soft, so overload stays feasible}$$

$$r_i \geq \text{floor}_i \quad \text{max-min floor — hard}$$

$$\sum_{i \in \sigma} \frac{r_i}{SE_i} + s_\sigma \geq \phi_\sigma C_d \quad \text{slice share — soft, demand-capped, work-conserving}$$

Solved in three stages, in that order:

$$\text{A} \quad t^* = \max t \quad \text{s.t.} \quad r_i \geq t \hat{G}_i \quad \forall i: c(i)=\text{GBR}, \quad \text{floor}_i = \alpha t^* \hat{G}_i$$

$$\text{B1} \quad S^* = \min \sum_i p_i s_i + p_\sigma \sum_\sigma SE_\sigma s_\sigma$$

$$\text{B2} \quad \max \sum_i w_{c(i)} \log(r_i + \epsilon) \quad \text{s.t.} \quad \sum_i p_i s_i \leq S^*$$

Order the objectives; do not weight

them. Folded into one, the two terms sit 3.3×10^7 apart — two solvers returned `optimal_inaccurate` and disagreed $3.6 \times$ on one flow's rate. Split into B1 then B2, both match the analytic optimum to 100 bps.

t^* is the largest fraction of contract every GBR flow can hold at once. $t^* = 1$

means the set is feasible, the floor binds nothing, and stage A is free — which is why it is on by default. α trades the guarantee against throughput.

Tier-2: hit that target, slot by slot

Every flow carries a virtual queue — an accumulator in bits, not a buffer of data:

$$\begin{aligned}
 Q_i &+= r_i \Delta && \text{grow by the Tier-1 target} \\
 Q_i &\leftarrow \min(Q_i, \text{ceil}_i) && \text{clamp to the legitimate debt} \\
 Q_i &\leftarrow \max(0, Q_i - \text{delivered}_i) && \text{serve, then drain} \\
 \text{ceil}_i &= \max(0, \min(r_i W, A_i^W) - D_i^W) && \text{arrived minus delivered over the window } W
 \end{aligned}$$

Ranking, and then the transport block:

$$\begin{aligned}
 \tilde{Q}_i &= Q_i + w_d \left(\frac{\text{HoL}_i}{\text{PDB}_i} \right)^\kappa \max_j Q_j && \text{urgency, scaled by the system-wide max queue} \\
 M_u &= \left(\sum_{i \in u} \tilde{Q}_i \right) \cdot \text{SE}_u && \text{rank UEs — opportunistic and rate-tracking at once}
 \end{aligned}$$

The granted transport block is then filled by TS 38.321 logical-channel prioritisation: by priority level, then by adjusted queue.

The virtual queue is a **non-saturating integrator**, so it can reach an arbitrary, unequal target vector — precisely what a multiplicative urgency metric cannot do at any weight.

Clamping to a **windowed arrival count**, not instantaneous backlog: a bursty video flow empties between frames, and clamping there erases its legitimate debt.

Configured grants carry the periodic flows — standing allocations in priority tiers, self-gating where they would starve the dynamic pool.

The simulator: what it models, and what it does not

Modelled

- The slot-by-slot PRB grid, with the TDD special slot
- The PDCCH/CCE budget, with DCI aggregation levels
- The uplink BSR round-trip — delay plus loss
- CQI delay and loss; BLER from MCS/SNR mismatch
- Per-UE AR(1) SNR through an MCS staircase
- Per-flow buffers, HoL timestamps, PDB expiry

Not modelled

- LDPC and I/Q; HARQ as a state machine — a fixed delay and a BLER discount instead
- Per-packet RLC segmentation, MU-MIMO
- Mobility, UE churn, inter-cell interference

So the results are **comparative, not absolute** — and what we omit would widen the configured-grant margin, not narrow it.

Three workloads, each sized so a different bottleneck binds:

Scenario	Carrier / TDD	Workload	What binds
factory_robots	40 MHz, $\mu=2$, DSUUU	10 robots, 24 flows, 14–24 dB	Uplink data channel
sensor_dense	30 MHz, $\mu=1$, DSUUU	30 periodic sensors, 15 ms PDB	PDCCH / CCE budget
latency_bound	40 MHz, $\mu=1$, DDDSU	8 × 5 Mbps + 80 Mbps bulk	Deadlines under saturation

Result 1: a capability PF does not have

30 dense periodic uplink sensors with a 15 ms deadline. The DCI/CCE budget runs out before the data channel does.

Scheduler	Total	On time	Min delivery	Worst p99
Round Robin	6.9 M	0/30	71%	15.0 ms
Proportional Fair	9.2 M	2/30	85%	15.0 ms
Two-tier	9.6 M	30/30	100%	5.0 ms

Each periodic flow gets a standing allocation that costs **zero PDCCH per slot and needs no BSR round trip**. PF is throttled by both budgets at once and has no mechanism to bypass either — so no amount of PF tuning closes this.

Larrañaga et al. (JNCA 2023) report the same dominance of configured grants for time-critical industrial traffic from ns-3 / 5G-LENA — an independent simulator sharing no code with ours.

Result 2: deadline-blindness is silent

Eight interactive 5 Mbps streams with a 12 ms deadline, sharing a saturated downlink with 80 Mbps of bulk.

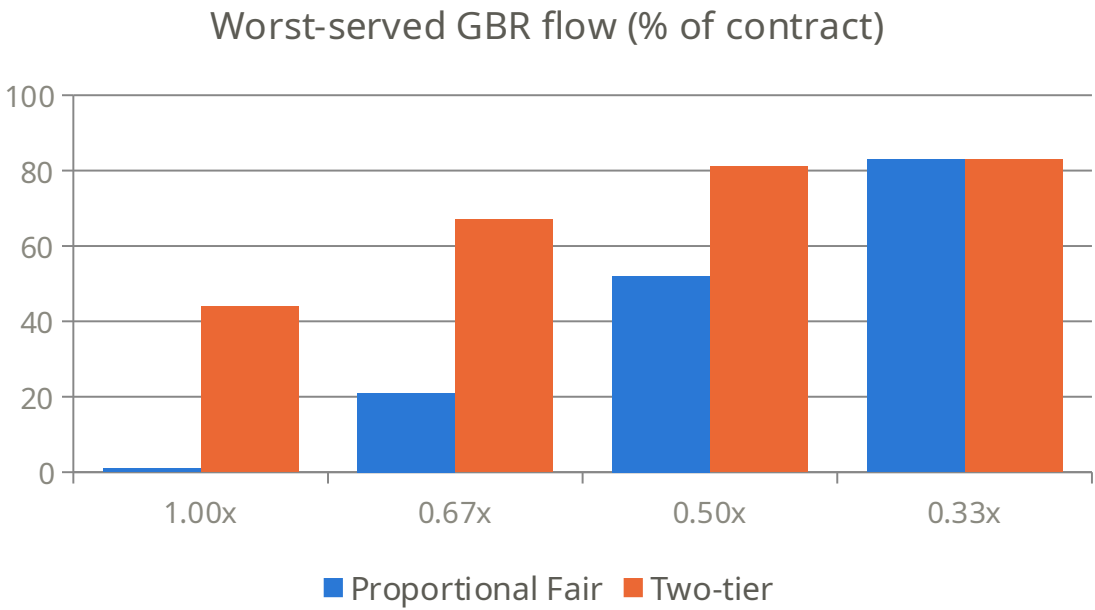
Scheduler	On time	Mean delivery	Worst p99	Bulk carried
Round Robin	3/8	84%	12.0 ms	22.8 M
Proportional Fair	5/8	86%	12.0 ms	24.6 M
Two-tier	8/8	100%	4.5 ms	13.2 M

PF's mean delivery on the control flows is **86% — which reads healthy on a dashboard**. The missing 14% are aged-out packets: the late motion commands, the safety-relevant ones. Mean delivery is not a safety metric.

The design pays for this explicitly and visibly — bulk throughput drops 24.6 → 13.2 Mbps to clear every deadline. That is the trade the QoS profile asked for.

Result 3: rate contracts, swept across offered load

Ten robots, uplink-heavy. Capacity is fixed by spectrum, so the swept axis is offered load.



Contracts met ($\geq 95\%$ of the rate floor)

Load	PF	Two-tier
1.00x (as shipped)	1/10	0/10
0.67x	6/10	1/10
0.50x	8/10	10/10
0.33x	10/10	10/10

At 0.50× we honour 10/10 contracts to PF's 8/10. Deeper in overload the two disagree about the metric, not the outcome: at 0.67× every robot holds $\geq 67\%$ of its rate where PF's worst gets 21%.

A threshold count rewards concentrating the shortfall; the max-min floor spreads it, and α picks between them. This deep into overload the operator's real lever is admission control.

What the design adds — and what it costs

Added over the PF baseline

- A **rate contract** that is stated and honoured — 10/10 against 8/10 at moderate load; worst-served 44% against 1% in overload.
- A deadline that is **tracked, not hoped for** — 8/8 against 5/8, p99 4.5 against 12.0 ms.
- A **control-channel path** for dense periodic traffic — 30/30 against 2/30 where the CCE budget binds.
- A **single fairness dial** that self-disables when the GBR set is feasible, plus soft per-slice floors.

Paid for with

- About **4% of aggregate throughput** at the max-min floor in deep overload — the price of holding the worst-served flow above zero.
- **One convex solve per second** — CVXPY/Clarabel here, GLPK simplex with successive convex approximation in the OAI port.
- Per-5QI priority and prioritised-bit-rate configuration that has to actually be set.

It is also **markedly less sensitive to control-plane fidelity**. Sweeping the uplink BSR round-trip from 0 to 16 slots, PF's contract count breaks 8/10 → 5/10 and its worst-served flow falls 66% → 30%; the two-tier design moves 10/10 → 8/10 and 80% → 77%, losing 1% of throughput against PF's 10%. Configured-grant flows are invariant to BSR delay by construction.

Where this stands, and what remains

Built and tested

- Scheduler library: Tier-1, Tier-2, configured grants, per-UE grants with a TS 38.321 multiplexer, per-slice floors
- A symbol-accurate NR simulator with PDCCH, BSR and CQI models
- Three scenario studies, 70 regression tests, one script per result table

In progress

- OAI integration in a separate repo — Tier-1 as GLPK simplex with successive convex approximation, Tier-2 inside the NR MAC
- Porting the max-min stage and the per-5QI priority mapping onto that branch

To test on hardware

- End-to-end runs on OAI with COTS UEs
- Tier-1 solve time inside a real 1 s budget
- What the simulator omits: HARQ PRB cost, uplink grant timing, mobility, UE churn
- Multi-cell and inter-cell interference

The design is settled and the simulation case is made: against the PF scheduler OAI ships, it adds rate contracts, deadline tracking and a control-channel path, at a cost we can state and bound. **The open work is landing it on real hardware** — everything above is comparative, in a simulator, and silicon is the next step.